

— FINAL PRESENTATION

Reppit

A Reddit-style forum with no application server and no database
— only a blockchain, IPFS, and a browser.

Solidity 0.8.28

React 19 + Vite

IPFS / Pinata

The Graph

Sepolia testnet

— THE PREMISE

Whoever runs the server owns the forum.

On a centralized forum, one company's database is the only copy of every post, vote, and ban. It can rewrite history, delete an account, or go offline — and the record goes with it.

Reppit puts the record somewhere nobody — including us — can quietly edit: **state lives in smart contracts, content lives on IPFS**, and the frontend is just a window onto both.

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application servers

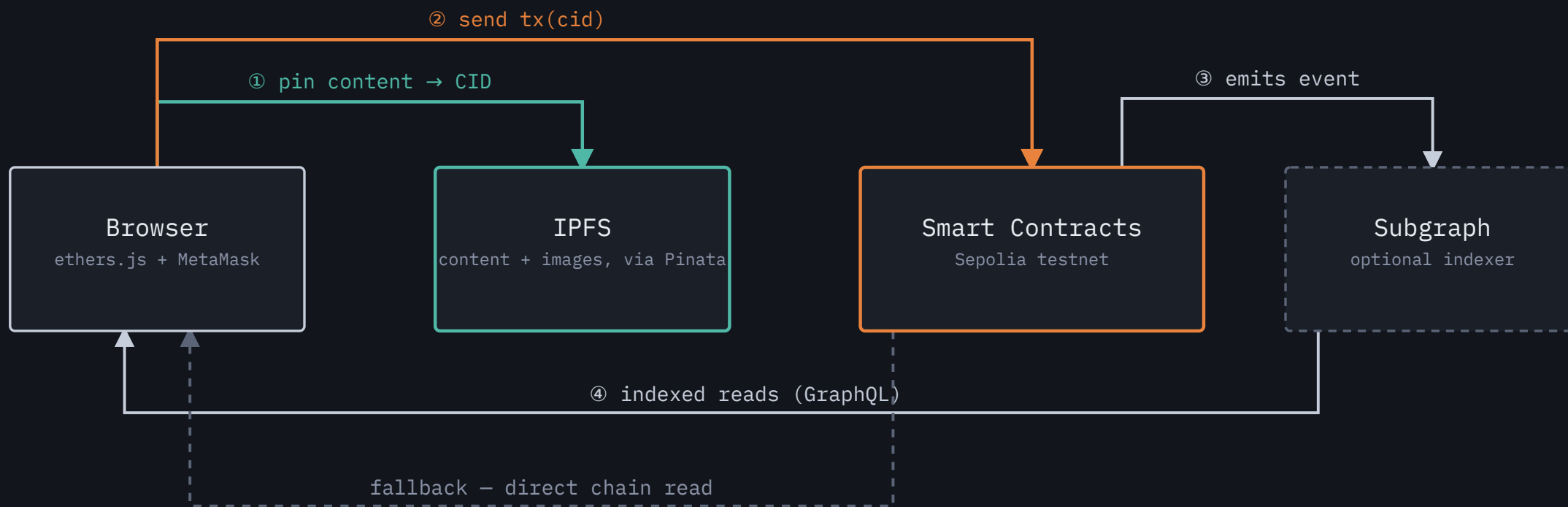
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central databases

3

independent smart contracts holding every fact

Three pieces, one shared truth.



Every write lands on-chain first; reads prefer the indexer for speed but fall back to reading the contracts directly when it's offline.

— on-chain — off-chain content (IPFS) — fallback path

— SOLIDITY

Split into three, on purpose.

mainContract.sol

DecentralizedForum

The core state-holder: communities, posts, votes, membership, activity scoring, and moderator governance.

viaIR: true

optimizer runs: 1

ForumModeration.sol

ForumModeration

Comments, the moderation queue, and report/resolve — split out purely to keep the main contract under the size limit.

≤ 24,576 B · EIP-170

UsernameRegistry.sol

UsernameRegistry

Standalone on-chain username registration — charset, length, uniqueness, and a rename cooldown.

independent of the other two

The two forum contracts call each other through a shared `IDecentralizedForum` / `IForumModeration` interface, wired together once at deploy time.

— PRODUCT

Communities, posts, votes.

COMMUNITIES

- Create or join an **r/name** community
- Membership gates posting, commenting, and voting

POSTS

- Up to **4 images**, **≤5 MB** each, pinned to IPFS
- Only the **CID** is written on-chain — never the file itself

VOTING

- **One vote per address** per post — same direction clears it, opposite direction switches it
- Comments carry no score, only posts do

— PRODUCT

Moderation is a vote, not a click.

— GAINING THE ROLE

- The **creator** is a permanent moderator
- The **two most-active members** — ranked by on-chain activity — are auto-promoted
- A moderator can **nominate** a member; once **3** moderators recommend them (or all of them, in communities with fewer than 3), the candidate must accept

— LOSING THE ROLE

- Any moderator except the creator can **resign**
- A **nominated** moderator can be removed by majority vote of the community's moderators, or the vote **expires after 7 days**
- A **ban** strips membership, role, and activity score — the creator can never be banned

A client-side safety filter routes flagged posts and comments into a **pending-review queue** instead of publishing them; moderators approve or reject from there, alongside member-filed reports.

— DEGRADES, NEVER BREAKS

The indexer is a cache, not a dependency.

Every subgraph query has a direct on-chain read behind it. Turn the indexer off and the app keeps working — some things just get slower or thinner.

FEATURE	WITH THE INDEXER	WITHOUT IT
Posts, comments, votes	fast GraphQL reads	direct chain reads — slower, same data
Trending feed	subgraph-ranked	reconstructed from chain reads
Search	full-text, cross-community	unavailable
Notifications	every event type	comment-reply notifications only
Profile stats & activity feed	full history	unavailable

— UNDER THE HOOD

The stack.

CHAIN

Solidity 0.8.28

Hardhat 3

ethers.js v6

STORAGE

IPFS

Pinata

FRONTEND

React 19

TypeScript

Vite

TanStack Query

INDEXING

The Graph

AssemblyScript

Docker

WALLET

MetaMask

— WHAT WENT WELL

Boundaries that let us split the work.

Independent packages

Contracts, frontend, and subgraph each own their own package .json — three teams could move in parallel without blocking each other.

Documentation quality

Hardhat, ethers.js, Solidity, and The Graph all have genuinely good docs — "how do we do X" was usually a docs search, not a dead end.

— MAIN DIFFICULTIES

Everything downstream of "it's not a web app."

- **Environment setup** — several npm installs, a Dockerized graph-node + Postgres + IPFS stack, and MetaMask network switching, all needing to land on the same working state per machine
- **A new mental model** — thinking in transactions, gas, on-chain vs. off-chain state, and wallet-signed calls, unlike anything the degree covered
- **Tools outside the curriculum** — Solidity, Hardhat, ethers.js, IPFS pinning, The Graph's AssemblyScript mappings, and Docker, each with its own learning curve

— WHAT'S NEXT

If we kept going.

- **WalletConnect** support, so the app isn't MetaMask-only
- An on-chain **reputation score** derived from voting history, shown on profiles
- Deploy to a **cheaper chain** to cut gas costs on everyday actions like voting and commenting
- Host the app and subgraph in the **cloud**, so using it doesn't start with a local setup

— THANK YOU

Questions?

reppit — a forum where the record is the chain.